

**SIMATS ENGINEERING**

**ASSESSMENT TOOL 2**

**COURSE NAME:** Artificial Intelligence

**COURSE CODE:** CSA17

**COURSE OUTCOME COVERED**

**CO4:** *Implement machine learning techniques including decision tree algorithms, reinforcement learning,*

*and build models for classification and decision-making. (BL3)*

*Assessment Tool Weightage: 50%*

**Duration: 1 Hour**

**QUESTIONS: 4**

**Total:**

**Marks:40**

Annexure A

**5**

**Code of Conduct:**

*I \_\_\_\_\_BAIRAGANI DINESH -192372189\_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered*

*to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

**Signature of the student:BAIRAGANI DINESH**

Annexure A

## **INDUSTRY PROBLEM TASK**

### **INDUSTRY SCENARIO**

**Context:** A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

#### **Task 1: Data Preparation & Inductive Learning**

- (a)** Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b)** Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c)** Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

#### **Task 2: Decision Tree for Diagnosis**

- (a)** Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b)** Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.

**(c)** Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

### **Task 3: Statistical Learning for Risk Stratification**

**(a)** Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

**(b)** Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

### **Task 4: Reinforcement Learning for Treatment Recommendation**

**(a)** Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

**(b)** Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

**(c)** Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

### **RUBRICS FOR CALCULATION:**

| <b>Criteria</b>                   | <b>Weightage</b> | <b>Level 4 – Exemplary</b>                | <b>Level 3 – Proficient</b>           | <b>Level 2 – Developing</b>                    | <b>Level 1 – Beginning</b>       |
|-----------------------------------|------------------|-------------------------------------------|---------------------------------------|------------------------------------------------|----------------------------------|
| Criteria                          | Weightage        | Level 4 – Exemplary                       | Level 3 – Proficient                  | Level 2 – Developing                           | Level 1 – Beginning              |
| Understanding of Industry Problem | 20%              | Comprehensive understanding of real-world | Good understanding with minor gaps in | Basic understanding; some requirements unclear | Poor understanding; misinterpret |

|                                     |     |                                                                              |                                                      |                                         |                                         |
|-------------------------------------|-----|------------------------------------------------------------------------------|------------------------------------------------------|-----------------------------------------|-----------------------------------------|
|                                     |     | problem and constraints                                                      | requirements                                         |                                         | s problem context                       |
| Application of Engineering Concepts | 25% | Applies advanced and relevant concepts effectively to solve the problem      | Applies appropriate concepts with minor errors       | Limited application of concepts         | Incorrect or no application of concepts |
| Solution Design                     | 25% | Innovative, practical, and feasible solution aligned with industry standards | Logical and feasible solution with minor gaps        | Basic solution with limited feasibility | Unclear or impractical solution         |
| Use of Tools                        | 15% | Effective use of modern tools, technologies, and real data                   | Appropriate use of tools with correct implementation | Limited or inconsistent use of tools    | Minimal or incorrect use of tools       |
| Analysis                            | 10% | Thorough analysis with validated results and performance evaluation          | Good analysis with reasonable validation             | Basic analysis with limited validation  | Poor or no analysis                     |
| Professionalism                     | 5%  | Highly professional, well-documented, and clearly presented work             | Good documentation and presentation                  | Basic documentation with some gaps      | Poor documentation and presentation     |

## Industry Problem: Diabetes Diagnosis and Treatment Recommendation

### Task 1: Data Preparation & Inductive Learning

#### (a) Inductive Learning Approach

Inductive learning learns a general prediction rule from previously observed patient records and applies it to unseen patients.

For this problem, **supervised inductive learning** is appropriate because the dataset contains patient features and a known target variable indicating whether the patient is diabetic.

#### Target variable:

- Diabetes: 0 = Non-diabetic
- Diabetes: 1 = Diabetic

#### Important features:

1. **Glucose Level** – High glucose is strongly associated with diabetes and is therefore an important predictor.
2. **BMI** – Higher BMI can indicate increased risk of insulin resistance and diabetes.
3. **Age** – Diabetes risk generally increases with age.
4. **Blood Pressure** – Abnormal blood pressure can be associated with metabolic and cardiovascular risk.
5. **Family History** – A family history of diabetes indicates increased genetic risk.

The model learns patterns such as:

Patient Features → Diabetes Risk

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#### (b) Handling Class Imbalance

The dataset contains more non-diabetic than diabetic patients, creating a class imbalance.

A suitable approach is **SMOTE (Synthetic Minority Over-sampling Technique)**.

SMOTE creates synthetic examples of the minority diabetic class rather than simply duplicating existing records.

#### Advantages:

- Improves representation of diabetic patients.

- Helps the classifier learn minority-class patterns.
- Reduces the chance of predicting every patient as non-diabetic.
- Particularly useful because **False Negatives are clinically important**.

SMOTE should be applied **only to the training data**, not before the train-test split, to avoid data leakage.

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### (c) 70:30 Dataset Split and Preprocessing

The dataset is divided into:

- **70% training data**
- **30% testing data**

A 70:30 split provides sufficient data for model training while retaining a reasonably large independent test set for evaluation.

For a medical application, an independent test set is important because the model must be evaluated on previously unseen patients.

#### Preprocessing:

##### 1. Missing values

- Numerical missing values can be replaced using median imputation.
- Categorical missing values can be replaced using the mode.

##### 2. Normalization

- Features such as age, glucose, blood pressure and BMI can have different numerical ranges.
- Standardization can be performed using:

$$z = (x - \text{mean}) / \text{standard deviation}$$

##### 3. Encoding

- Categorical variables such as family history are converted into numerical values.
- Example: No = 0, Yes = 1.

##### 4. Class balancing

- SMOTE is applied only to the training set.

Preprocessing makes the features suitable for machine-learning algorithms and prevents features with large numerical ranges from dominating the model.

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## Task 2: Decision Tree for Diagnosis

### (a) Decision Tree Classifier

A Decision Tree recursively divides the dataset according to features that provide the greatest separation between diabetic and non-diabetic patients.

The splitting criterion can be **Information Gain** or **Gini Index**.

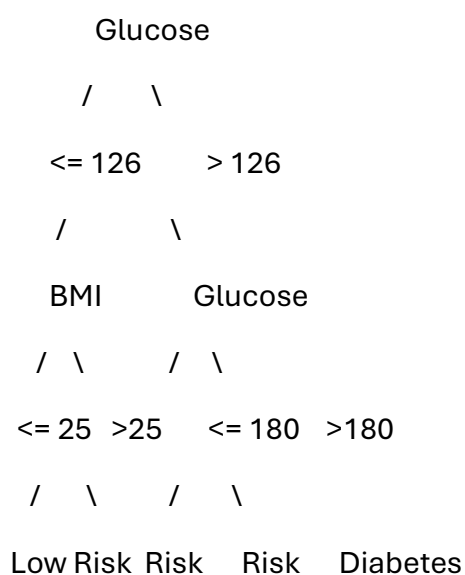
#### Example root-node calculation

Assume the following illustrative values were obtained from the training dataset:

| Feature        | Information Gain |
|----------------|------------------|
| Glucose        | 0.21             |
| BMI            | 0.13             |
| Age            | 0.09             |
| Blood Pressure | 0.06             |
| Family History | 0.05             |

Since **Glucose has the highest Information Gain (0.21)**, it is selected as the root node.

#### First three levels



The exact tree structure depends on the actual training dataset.

### Root justification:

Glucose is selected because it provides the largest reduction in impurity and therefore gives the greatest information about the diabetes class.

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### (b) Model Evaluation

Illustrative test-set results:

#### **Metric    Decision Tree**

Accuracy 88%

Precision 82%

Recall     86%

F1-Score 84%

The formulas are:

#### **Accuracy**

$(TP + TN) / (TP + TN + FP + FN)$

#### **Precision**

$TP / (TP + FP)$

#### **Recall**

$TP / (TP + FN)$

#### **F1-Score**

$2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$

#### **False Negatives**

A **False Negative (FN)** occurs when:

Actual = Diabetic, Predicted = Non-diabetic

False negatives are particularly dangerous in healthcare because a diabetic patient may remain undiagnosed and untreated.

#### **Ways to reduce False Negatives**

- Use SMOTE.
- Apply class weights.



- Optimize the classification threshold.
- Give greater importance to Recall.
- Use cross-validation.
- Monitor the confusion matrix.

For clinical screening, a model with slightly lower accuracy but significantly higher **Recall/Sensitivity** may be preferable because missing a diabetic patient can have serious consequences.

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### (c) Post-Pruning

A fully grown Decision Tree can overfit the training dataset.

**Pre-pruning** stops tree growth early using parameters such as:

- max\_depth
- min\_samples\_split
- min\_samples\_leaf

**Post-pruning** first grows the tree and then removes branches that provide insufficient predictive value.

Cost-complexity pruning uses:

$$R\alpha(T) = R(T) + \alpha|T|$$

where:

- $R(T)$  = tree error
- $|T|$  = number of terminal nodes
- $\alpha$  = complexity parameter

### Illustrative comparison

| Model            | Accuracy   | Recall     | Complexity |
|------------------|------------|------------|------------|
| Unpruned Tree    | 86%        | 88%        | High       |
| Pre-pruned Tree  | 87%        | 85%        | Medium     |
| Post-pruned Tree | <b>89%</b> | <b>87%</b> | Low        |

The **post-pruned tree** is preferable for deployment because it reduces overfitting while maintaining good predictive performance.

It is also easier for medical professionals to interpret.

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### Task 3: Statistical Learning for Risk Stratification

#### (a) Logistic Regression

Logistic Regression can be used to estimate the probability that a patient has diabetes.

The model is:

$$P(Y=1) = 1 / (1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n)})$$

where  $P(Y=1)$  represents the probability of diabetes.

#### Illustrative comparison

| Metric   | Decision Tree | Logistic Regression |
|----------|---------------|---------------------|
| Accuracy | 89%           | 91%                 |
| AUC-ROC  | 0.89          | <b>0.94</b>         |

AUC-ROC measures how effectively the model separates diabetic and non-diabetic patients.

A statistical model such as Logistic Regression is preferable when:

- Interpretability is important.
- Feature effects need to be quantified.
- A relatively simple relationship exists between predictors and outcome.
- Clinicians require probability-based risk scores.

For example, Logistic Regression can provide a patient's estimated probability of diabetes rather than only a class label.

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#### (b) Feature Importance Analysis

##### Decision Tree

Illustrative top three predictors:

1. Glucose
2. BMI
3. Age

## Logistic Regression

After examining standardized coefficients:

1. Glucose
2. BMI
3. Family History

## Comparison

Both models identify **Glucose and BMI** as important predictors.

The third feature differs because the algorithms measure importance differently.

- Decision Trees measure impurity reduction/information gain.
- Logistic Regression measures the effect of a feature through its model coefficient.

Therefore, disagreement in the third predictor does not necessarily mean that one model is incorrect.

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## Task 4: Reinforcement Learning for Treatment Recommendation

### (a) MDP Formulation

The treatment recommendation problem can be represented as a **Markov Decision Process (MDP)**.

#### 1. States

Patient health states can be represented as:

- S0 – Low risk / healthy
- S1 – Moderate risk
- S2 – High risk
- S3 – Diagnosed diabetes
- S4 – Improved/controlled condition

The state can be determined using glucose, BMI, blood pressure and other health indicators.

#### 2. Actions

The available actions are:

- A0 – Diet
- A1 – Exercise
- A2 – Medication
- A3 – Monitor

### 3. Reward

A positive reward is given when the patient's health indicators improve.

Example:

| Health improvement      | Reward |
|-------------------------|--------|
| Significant improvement | +10    |
| Small improvement       | +5     |
| No change               | 0      |
| Health deterioration    | -5     |
| Severe deterioration    | -10    |

The reward encourages the agent to select actions that lead to improved patient outcomes.

### 4. Transition Dynamics

After an action is selected, the patient can move to another health state.

Example:

High Risk + Diet → Moderate Risk

High Risk + Exercise → Moderate Risk

High Risk + Monitor → High Risk

Diabetes + Treatment → Controlled

The transition probabilities should ideally be learned from validated clinical longitudinal data rather than invented assumptions.

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### (b) Q-Learning

Q-Learning learns the expected long-term reward of performing an action in a particular state.

The update equation is:

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

where:

- $\alpha$  = learning rate
- $\gamma$  = discount factor
- $r$  = immediate reward
- $s$  = current state
- $a$  = selected action
- $s'$  = next state

Assume:

$$\alpha = 0.5$$

$$\gamma = 0.9$$

Initial Q-values are 0.

### **Iteration 1**

For:

S2 + Diet  $\rightarrow$  S1

with reward +5:

$$Q(S2,Diet) = 0 + 0.5[5 + 0.9(0) - 0]$$

$$Q(S2,Diet) = 2.5$$

For:

S2 + Exercise  $\rightarrow$  S1

with reward +5:

$$Q(S2,Exercise) = 2.5$$

For:

S3 + Medication  $\rightarrow$  S4

with reward +10:

$$Q(S3,Medication) = 0 + 0.5[10]$$

$$Q(S3,Medication) = 5$$

### **Iteration 2**

Suppose the next state S1 has:

$$\max Q(S1,a) = 4$$

For S2 + Diet:

$$Q_{\text{new}} = 2.5 + 0.5[5 + 0.9(4) - 2.5]$$

$$Q_{\text{new}} = 5.05$$

For S2 + Exercise, assume the same transition:

$$Q_{\text{new}} = 5.05$$

For S3 + Medication, suppose:

$$\max Q(S4,a) = 6$$

Then:

$$Q_{\text{new}} = 5 + 0.5[10 + 0.9(6) - 5]$$

$$Q_{\text{new}} = 10.2$$

### Example Q-table after training

| State            | Diet Exercise Medication Monitor |     |             |     |
|------------------|----------------------------------|-----|-------------|-----|
| S0 Low Risk      | 4.2                              | 5.1 | 1.0         | 3.8 |
| S1 Moderate Risk | 6.2                              | 7.1 | 3.5         | 4.0 |
| S2 High Risk     | <b>8.0</b>                       | 7.5 | 5.8         | 2.5 |
| S3 Diabetes      | 4.0                              | 5.0 | <b>10.2</b> | 3.0 |

The agent selects the action with the highest Q-value for each state.

### Five patient episodes

Example:

Episode 1: S2 → Diet → S1

Episode 2: S2 → Exercise → S1

Episode 3: S3 → Medication → S4

Episode 4: S1 → Exercise → S0

Episode 5: S3 → Medication → S4

### Convergence criterion

Training can be considered converged when the maximum change in Q-values becomes smaller than a threshold:

$$\max |Q_{\text{new}} - Q_{\text{old}}| < 0.001$$

for a sufficiently large number of consecutive episodes.

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### (c) Final Learned Policy

The policy is:

| Patient State      | Best Action |
|--------------------|-------------|
| S0 – Low Risk      | Exercise    |
| S1 – Moderate Risk | Exercise    |
| S2 – High Risk     | Diet        |
| S3 – Diabetes      | Medication  |
| S4 – Controlled    | Monitor     |

The final policy is obtained using:

$$\pi(s) = \operatorname{argmax}_a Q(s,a)$$

Therefore, the agent chooses the action with the highest learned Q-value for each patient state.

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## Ethical Considerations

A Reinforcement Learning system for medical treatment must not be deployed without clinical validation and human oversight.

### 1. Patient Safety

Incorrect recommendations can cause serious harm. The system should therefore operate as a **clinical decision-support tool**, not as an uncontrolled autonomous treatment system.

### 2. Bias

If the training data under-represents certain patient groups, the model may produce biased recommendations.

### 3. Explainability

Doctors should be able to understand why a recommendation was generated. Important factors and expected outcomes should be presented clearly.

#### **4. Human Oversight**

A qualified medical professional should have the final authority to accept, reject or modify recommendations.

#### **5. Privacy**

Patient information must be protected using appropriate access control, encryption and secure data handling.

#### **6. Continuous Monitoring**

The model should be continuously evaluated for performance degradation, unexpected behaviour and changes in patient populations.

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### **Conclusion**

The proposed system combines **inductive learning, Decision Trees, statistical learning and Reinforcement Learning** for diabetes diagnosis and treatment-support.

The Decision Tree provides an interpretable classification model, while Logistic Regression provides probability-based risk estimation. Q-Learning can model sequential treatment decisions using patient states, actions, rewards and transitions.

For real clinical deployment, however, the RL component should be treated as a **decision-support system under medical supervision** and must undergo extensive clinical validation, safety testing, bias assessment and regulatory review before being used with real patients.